

Diffie-Hellman Key Exchange

Course Title: Information Security Laboratory

Course No: 0714 02 CSE 4108

Submitted to,

Dr. Anupam Kumar Bairagi
Professor,
Computer Science and Engineering Discipline,
Khulna University, Khulna-9208

Submitted by,

Md. Siam Ahmed
Student ID: 230202
Year: 4th, Term: 1st
Computer Science and Engineering Discipline
Khulna University, Khulna-9208

The Implementation of Diffie-Hellman Key Exchange:

```
#include <iostream>
#include <cstdlib>

using namespace std;
#define int long long

int bin_exp(int a, int b, int mod)
{
    if (b == 0)
        return 1;

    int val = bin_exp(a, b / 2, mod);

    if (b % 2 == 0)
        return (1LL * val * val) % mod;
    else
        return (((1LL * val * val) % mod) * a) % mod;
}

int32_t main()
{
    cout << "Diffie Hellman Key Exchange:\n";
    int q, a, xa, xb;
    cout << "Enter the value of q, a, XA, and XB respectively: ";
    cin >> q >> a >> xa >> xb;
    int ya, yb, ka, kb;
    cout << "Global Public Elements:\n";
    cout << "\tPrime Number(q): " << q << endl;
    cout << "\tPrimitive Root(a): " << a << endl;
    cout << "Private Key:\n";
    cout << "\t" << "XA: " << xa << endl;
    cout << "\tXB: " << xb << endl;
    cout << "Public Key:\n";
    cout << "\tYA: " << (ya = bin_exp(a, xa, q)) << endl;
    cout << "\tYB: " << (yb = bin_exp(a, xb, q)) << endl;
    cout << "Secrekt Key:\n";
    cout << "\tKA: " << (ka = bin_exp(yb, xa, q)) << endl;
    cout << "\tKB: " << (kb = bin_exp(ya, xb, q)) << endl;
    cout << "The secret key of both user are " << (ka == kb ? "matched." : "not
matched.") << endl;
    return 0;
}
```

Input/Output:

```
Diffie Hellman Key Exchange:
Enter the value of q, a, XA, and XB respectively: 1000000009 14142135 874521963 932145876
Global Public Elements:
    Prime Number(q): 1000000009
    Primitive Root(a): 14142135
Private Key:
    XA: 874521963
    XB: 932145876
Public Key:
    YA: 974000290
    YB: 521113455
Secret Key:
    KA: 819304699
    KB: 819304699
The secret key of both user are matched.
```